

Hands-on Lab : Simple SELECT Statements

Estimated time needed: 20 minutes

In this lab, you will learn one of the most commonly used statements of SQL (Structured Query Language), the SELECT statement. The SELECT statement is used to select data from a database.

Objectives

After completing this lab, you will be able to:

- Query a database
- Retrieve data records from one or more tables of a database as resultset according to the criteria you specify

Concepts covered

How does the syntax of a SELECT statement look?

```
SELECT column1, column2, ...  
FROM table_name  
WHERE condition  
;
```

What do the keywords / clauses of a SQL statement shown above do?

- **FROM:** Specifies from which table to get the data. The clause can include optional JOIN subclauses to specify the rules for joining tables.
- [Optional Clause] **WHERE :** Specifies which rows to retrieve.

Why is there a semicolon after the SQL statements?

- Some database systems require a semicolon at the end of each SQL statement for execution. It is a standard way to separate one SQL statement from another which allows more than one SQL statement to be executed in the same call to the server. So, it is good practice to use a semicolon at the end of each SQL statement.

Introduction to Lab Environment

Software Used in this Lab

In this lab, you will use Datasette, an open source tool for exploring and publishing data. You can visit the [Datasette GitHub repository here](#).

Working with Datasette

The **Datasette** tool offers a platform to input and execute SQL queries. By clicking the **Submit query** button, you can execute the SQL query.

home / SanFranciscoFilmLocations

Practice SQL

Database: SanFranciscoFilmLocations

SELECT * FROM FilmLocations;

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

Support/Feedback

Powered by [Datasette](#)

Database Used in this Lab

The database used in this lab comes from the following dataset source: [Film Locations in San Francisco](#) under a [PDDL: Public Domain Dedication and License](#).

Task A: Exploring the Database

Let us first explore the **SanFranciscoFilmLocations** database using the **Datasette** tool:

1. If the first statement listed below is not already in the Datasette textbox on the right, then copy the code below by clicking on the little copy button on the bottom right of the codeblock below and then paste it into the textbox of the Datasette tool using either **Ctrl+V** or right-click in the text box and choose **Paste**.

```
SELECT * FROM FilmLocations;
```

home / Practice SQL / SanFranciscoFilmLocations

Practice SQL

Database: SanFranciscoFilmLocations

1 SELECT * FROM FilmLocations;

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

2. Click **Submit Query**.
3. Now you can scroll down the table and explore all the columns and rows of the **FilmLocations** table to get an overall idea of the table contents.

Title	ReleaseYear	Locations	FunFacts	ProductionCompany	Distributor	Director	Writer	Actor1	Actor2	Actor3
180	2011	Epic Roasthouse (399 Embarcadero)		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Srimam, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	Mason & California Streets (Nob Hill)		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Srimam, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	Justin Herman Plaza		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Srimam, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	200 block Market Street		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Srimam, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	City Hall		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Srimam, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	Pulk & Larkin Streets		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Srimam, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	Randall Museum		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Srimam, & Suba	Siddarth	Nithya Menon	Priya Anand

4. These are the column attribute descriptions from the **FilmLocations** table:

```
FilmLocations(  
  Title:           titles of the films,  
  ReleaseYear:     time of public release of the films,  
  Locations:       locations of San Francisco where the films were shot,  
  FunFacts:       funny facts about the filming locations,  
  ProductionCompany: companies who produced the films,  
  Distributor:     companies who distributed the films,  
  Director:       people who directed the films,  
  Writer:         people who wrote the films,  
  Actor1:         person 1 who acted in the films,  
  Actor2:         person 2 who acted in the films,  
  Actor3:         person 3 who acted in the films  
)
```

Task B: Example exercises on SELECT statement

Now let us go through some examples of SELECT queries:

1. In this example, suppose we want to retrieve details of all the films from the **FilmLocations** table. The details of each film record should contain all the film columns.

1. Problem:

*Retrieve all records with all columns from the **FilmLocations** table.*

2. Solution:

```
SELECT * FROM FilmLocations;
```

3. Copy the solution code above by clicking on the little copy button on the bottom right of the codeblock below and paste it to the textbox of the Datasette tool. Then click **Submit query**.

home / Practice SQL / SanFranciscoFilmLocations

Practice SQL

Database: SanFranciscoFilmLocations

1

SELECT * FROM FilmLocations;

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

4. Your output resultset should match like below:

Database: SanFranciscoFilmLocations

1

SELECT * FROM FilmLocations;

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

Results

All commands ran successfully

SELECT * FROM FilmLocations

Title	ReleaseYear	Locations	FunFacts	ProductionCompany	Distributor	Director	Writer	Actor1	Actor2	Actor3
180	2011	Epic Roasthouse (390 Embarcadero)		SPI Cinemas		Jayendra	Umargi Anuradha, Jayendra, Aarthi Sriram, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	Mason & California Streets (Nob Hill)		SPI Cinemas		Jayendra	Umargi Anuradha, Jayendra, Aarthi Sriram, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	Justin Herman Plaza		SPI Cinemas		Jayendra	Umargi Anuradha, Jayendra, Aarthi Sriram, & Suba	Siddarth	Nithya Menon	Priya Anand

2. In this example, now we want to retrieve selective details of all the film records. Let us retrieve the names of all the films along with director names and writer names.

1. Problem:

Retrieve the names of all films with director names and writer names.

2. Solution:

SELECT Title, Director, Writer FROM FilmLocations;

3. Copy the solution code above by clicking on the little copy button on the bottom right of the codeblock below and paste it to the textbox of the Datasette tool. Then click **Submit query**.

home / Practice SQL / SanFranciscoFilmLocations

Practice SQL

Database: SanFranciscoFilmLocations

1

SELECT Title, Director, Writer FROM FilmLocations;

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

4. Your output resultset should match like below:

Practice SQL

Database: SanFranciscoFilmLocations

1 SELECT Title, Director, Writer FROM FilmLocations;

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

Results

All commands ran successfully

SELECT Title, Director, Writer FROM FilmLocations

Title	Director	Writer
180	Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba
180	Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba
180	Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba
180	Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba
180	Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba
180	Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba
180	Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba

3. In this example, we want to retrieve film names along with filming locations and release years. But we also want to restrict the output resultset so that we can retrieve only the film records released in 2001 and onwards (release years after 2001 including 2001).

1. Problem:

Retrieve the names of all films released in the 21st century and onwards (release years after 2001 including 2001), along with filming locations and release years.

2. Solution:

SELECT Title, ReleaseYear, Locations FROM FilmLocations WHERE ReleaseYear>=2001;

3. Copy the solution code above by clicking on the little copy button on the bottom right of the codeblock below and paste it to the textbox of the Datasette tool. Then click **Submit query**.

home / Practice SQL / SanFranciscoFilmLocations

Practice SQL

Database: SanFranciscoFilmLocations

1 SELECT Title, ReleaseYear, Locations FROM FilmLocations WHERE ReleaseYear>=2001;

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

4. Your output resultset should match like below:

Practice SQL

Database: SanFranciscoFilmLocations

1 SELECT Title, ReleaseYear, Locations FROM FilmLocations WHERE ReleaseYear>=2001;

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

Results

All commands ran successfully

SELECT Title, ReleaseYear, Locations FROM FilmLocations WHERE ReleaseYear>=2001

Title	ReleaseYear	Locations
180	2011	Epic Roasthouse (399 Embarcadero)
180	2011	Mason & California Streets (Nob Hill)
180	2011	Justin Herman Plaza
180	2011	200 block Market Street
180	2011	City Hall
180	2011	Polk & Larkin Streets
180	2011	Randall Museum
180	2011	555 Market St.
24 Hours on Craigslist	2005	
About a Boy	2014	Broderick from Fulton to McAllister

Task C: Practice exercises on SELECT statement

Finally, let us practice creating and running some SELECT queries.

1. Problem:

Retrieve the fun facts and filming locations of all films.

- ▶ [Click here for Hint](#)
- ▶ [Click here for Solution](#)
- ▶ [Click here for Output](#)

2. Problem:

Retrieve the names of all films released in the 20th century and before (release years before 2000 including 2000) that, along with filming locations and release years.

- ▶ [Click here for Hint](#)
- ▶ [Click here for Solution](#)
- ▶ [Click here for Output](#)

3. Problem:

Retrieve the names, production company names, filming locations, and release years of the films which are not written by James Cameron.

- ▶ [Click here for Hint](#)
- ▶ [Click here for Solution](#)
- ▶ [Click here for Output](#)

Thank you for completing this lab where you learnt to query a database, and retrieve data records according to the criteria specified!

Author(s)

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